

Toronto Rush Hour Traffic Analysis*

INF2167 Project - Group 7

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*Code and data are available at: https://github.com/srg-uoft/INF2167_Group7_Project.

1 Introduction

Stella: I think the goal for this is a brief 1-2 sentences to give an overview of the project

2 Research Questions(s) & Hypothesis

This project will aim to address some research questions and preeminent assumptions about traffic patterns, as discussed below.

2.1 Questions

Stella's Rough Notes:

Note: mention using a data first approach, finding an interesting and up to date data set and determining what questions can be answered with it

What are the most accident-prone locations and times for travellers in the GTA, and how does that vary depending on mode of transportation?

Does the GTA's data align with pre-existing research on riskiest travel times?

2.2 Hypotheses

Stella's Rough Notes:

GTA traffic risk is similar to pre-established riskiest days for accidents

Some hypotheses we have are...

These are of interest because...

3 Relevant Literature

Stella's Ideas For Potential Sources

Days that have statistically the most car accidents

Mortality rates of various transportation methods

Toronto statistically has the worst traffic

congestion in Canadian cities not only costs time and sacrifices productivities, but also increases environmental burdens by more greenhouse gas emission. (<https://comt.ca/Reports/High%20Cost%20of%20Congestion%202012.pdf>)

4 Dataset & Methods

4.1 Dataset

4.1.1 Source

Dataset obtained from [CITE TPS HERE].

4.1.2 Structure

This is how our dataset is structured

4.1.3 Key Variables

Key variables from our dataset are...

4.2 Methods

Methods we plan to apply in our analysis are...

5 Exploratory Data Analysis

5.1 Overview

Initial exploratory data analysis was done using statistical programming language (R Core Team 2023) and the Tidyverse library (Wickham et al. 2019), with the goal of uncovering high-level trends and possible variables of interest for further analysis. Key discoveries which we hope will aid in answering our research question are elaborated on below.

5.2 Key Aspect 1

One key dataset aspect that relates to our variables of interest is...

5.3 Key Aspect 2

Another key dataset aspect that relates to our variables of interest is...

5.4 Key Aspect 3

Another key dataset aspect that relates to our variables of interest is...

6 Next Steps

Having outlined the research question of interest, target data set, and preliminary exploratory data analysis above, we will move into more detailed and thorough analysis of the dataset to identify any trends which could help address the primary research question. Conclusions and discussion, along with raw results and any models used in their generation, will be presented in a final report and oral presentation in August 2026.

7 References

- R Core Team. 2023. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. “Welcome to the Tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.